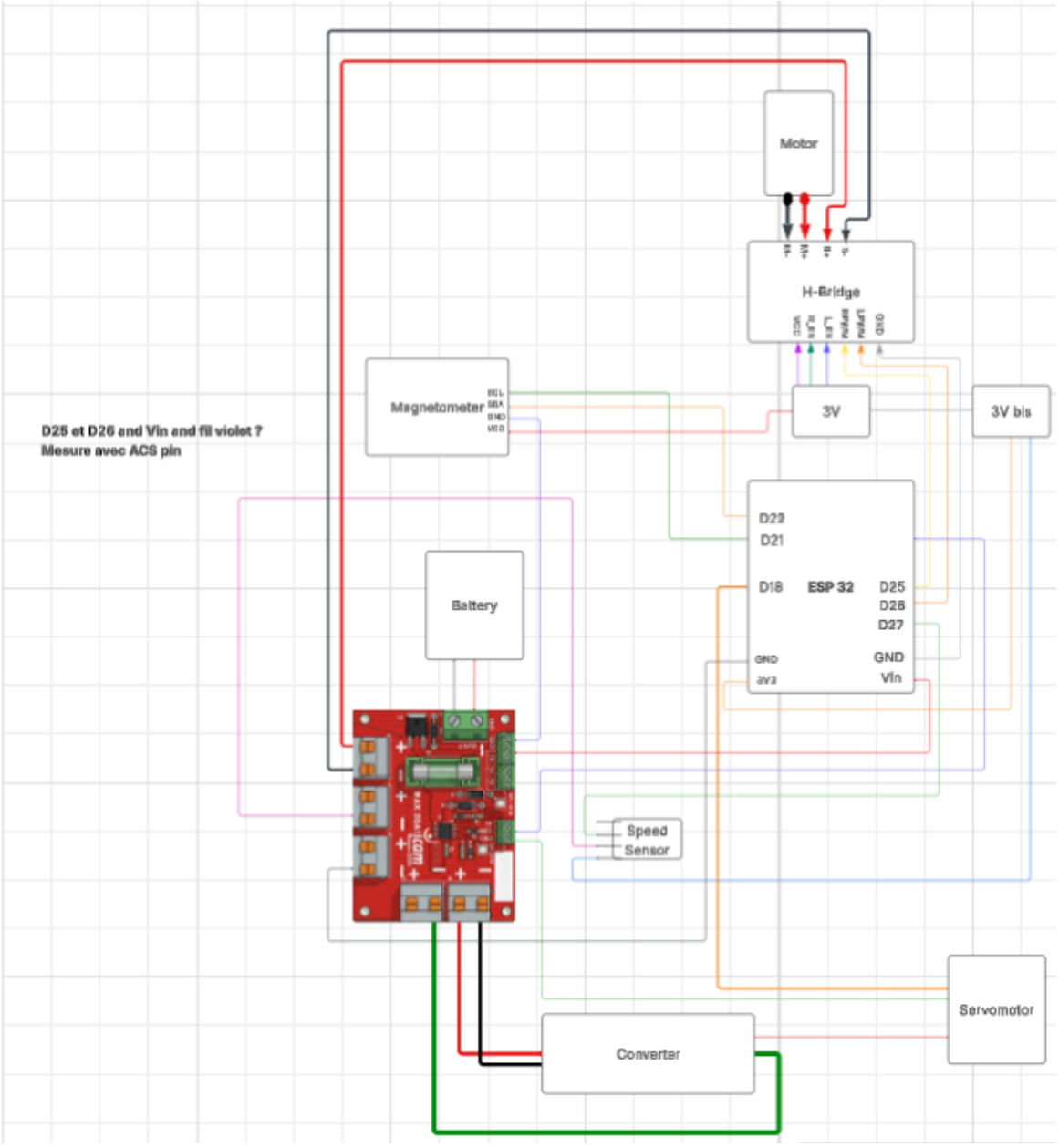
INDIVIDUAL TEST BENCH TEST BOAT PROJECT REPORT MIA & EEE

1. Introduction

- **Project objective**

The objective is to build an autonomous controllable boat integrating a wifi communication system functional. Monitoring the system step by step.

- **Scope of the system**



- **Brief overview of main components (ESP32, sensors, actuators)**

We have the main board ESP32, H-bridge, Current card, sensors(current and speed), DC-DC converter, Dc Motor, Servo motor, Magnetometer and wires connection

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2. System Overview

From the Scope overview (Above) you have the general block diagram of the system.

- **Description of how components interact:**
    - **ESP32 (Main Controller)**

Runs everything: WiFi server + control logic + sensor processing, Generates PWM signals for motor + servo, Reads sensors (I2C, analog, interrupts), Hosts web interface (manual + autopilot)

    - **Servo motor for steering**

Controls rudder direction (left/right steering), Receives angle (0–180°), Used by joystick + autopilot PID

    - **H-bridge for motor control**

- Controls DC motor direction: Forward (RPWM) and Reverse (LPWM)
- Controls speed via PWM duty cycle

    - **Magnetometer (BMM150)**

Measures Earth's magnetic field, Calculates boat heading (0–360°), Used for autopilot correction

    - **Speed Sensor (pulse interrupt)**

Counts rotation pulses from propeller or Hall sensor, Calculates RPM + speed.

    - **Current Sensor (ACS-style analog sensor**

Measures motor consumption (Amps), Detects load / resistance / stall

    - **DC-DC Converter (Power to servo)**

Converts battery voltage stable 5V/6V, Powers servo independently (important!), Prevents ESP32 brownout

3. Hardware Description (Test Bench Description)

3.1 Electrical Connections

- **Wiring diagram** ; Each component is wired from the ESP32 board to the other components, most having GND and 3.3V connections. Except the servo motor which needs an alimentation of 5V given by the output of the DC-DC converter.
- **Power distribution explanation ;** The whole system is alimented by a 12V battery passing through the current card then alimenting other components of the whole circuit.

3.2 Test Bench Setup

- Tools used (power supply, multimeter)

  Multimeter is used to ensure the is normal flow of voltage in the circuit and ensure Good communication of all Grounds(GND) and 5V

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4. Software Description

- Development environment: **Arduino IDE**
- Structure of the program:
  - Initialization
  - Sensor reading
  - Control logic
  - Actuator commands
- Communication protocols (I2C, analog, PWM)

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5. Test Procedures (Test Sheets)

For each test,:

Test 1: SERVO TEST (Steering)

- **Objective:** To control rudder direction
- **Expected Result:** Shows steering range from 0°-180°

Test 2: H-BRIDGE MOTOR TEST (BTS7960)

- **Objective:** To control Dc Motor Speed and direction

CODES;

```
#include <ESP32Servo.h>

Servo testServo;
const int servoPin = 18;

void setup() {
  Serial.begin(115200);
  testServo.attach(servoPin, 500, 2400);
}

void loop() {
  Serial.println("0°");
  testServo.write(0);
  delay(1000);

  Serial.println("90°");
  testServo.write(90);
  delay(1000);

  Serial.println("180°");
  testServo.write(180);
  delay(1000);
}
```

```
#define RPWM 25
#define LPWM 26

#define CHANNEL_R 0
#define CHANNEL_L 1

void setup() {
  Serial.begin(115200);

  ledcAttach(RPWM, 20000, 8);
  ledcAttach(LPWM, 20000, 8);
}

void loop() {
  Serial.println("Forward");
  ledcWrite(RPWM, 150);
  ledcWrite(LPWM, 0);
  delay(2000);

  Serial.println("Stop");
  ledcWrite(RPWM, 0);
  ledcWrite(LPWM, 0);
  delay(1000);
```

Test 3: MAGNETOMETER TEST (BMM150)

- **Objective:** Give the actual heading values of the boat

Test 4: SPEED SENSOR TEST (RPM)

- **Objective:** Gives the actual speed rotation of the motor

CODES;

```
#include <Wire.h>
#include <bmm150.h>

BMM150 bmm;

void setup() {
  Serial.begin(115200);
  Wire.begin(22, 21);

  if (bmm.initialize() == BMM150_OK) {
    Serial.println("BMM150 OK");
  } else {
    Serial.println("BMM150 FAIL");
  }
}

void loop() {
  bmm.read_mag_data();

  int x = bmm.raw_mag_data.raw_datax;
  int y = bmm.raw_mag_data.raw_datay;

  float heading = atan2(y, x) * 180 / PI;
```

```
#define SENSOR_PIN 27

volatile int pulses = 0;

void IRAM_ATTR countPulse() {
  pulses++;
}

void setup() {
  Serial.begin(115200);
  pinMode(SENSOR_PIN, INPUT);
  attachInterrupt(digitalPinToInterrupt(SENSOR_PIN), countPulse, FALLING);
}

void loop() {
  delay(1000);

  int rpm = pulses * 60; // 1 pulse per rotation assumed

  Serial.print("Pulses: ");
  Serial.print(pulses);
  Serial.print(" | RPM: ");
  Serial.println(rpm);
```

Test 5: CURRENT SENSOR TEST

- **Objective:** Calculate the actual current in the circuit

```
#define PIN_MES 34

float Vref = 3.3;
int ADCmax = 4095;
float sensitivity = 0.04;

void setup() {
  Serial.begin(115200);
}

void loop() {
  int raw = analogRead(PIN_MES);
  float voltage = (raw * Vref) / ADCmax;

  float current = (voltage - 1.65) / sensitivity;

  Serial.print("Voltage: ");
  Serial.print(voltage);
  Serial.print(" V | Current: ");
  Serial.print(current);
  Serial.println(" A");

  delay(500);
```

5.1 Network Configuration

5.1.1 Wi-Fi Access Point Setup

- **Objective:** Enable wireless communication between devices
- The ESP32 was configured as a Wi-Fi Access Point (AP)
- **Parameters:**
  - SSID: `RedmiNote14`
  - Password: `12345olfa`
  - IP Address: Typically 192.168.4.1, If using the arduino Wifi
- **Result:**
  Other devices successfully connect to the ESP32 network.

1.2 Client-Server Architecture

- **Server:** ESP32 hosting a web server (`#include <WiFi.h>`)
- **Client:** Another ESP32 or external device sending requests or creating our own wifi communication with the ESP32 Board.

```
const char* STA_SSID = "RedmiNote14";

const char* STA_PASSWORD = "12345olfa";
```

OR

```
const char* AP_SSID = "BoatControl";

const char* AP_PASSWORD = "12345678";
```

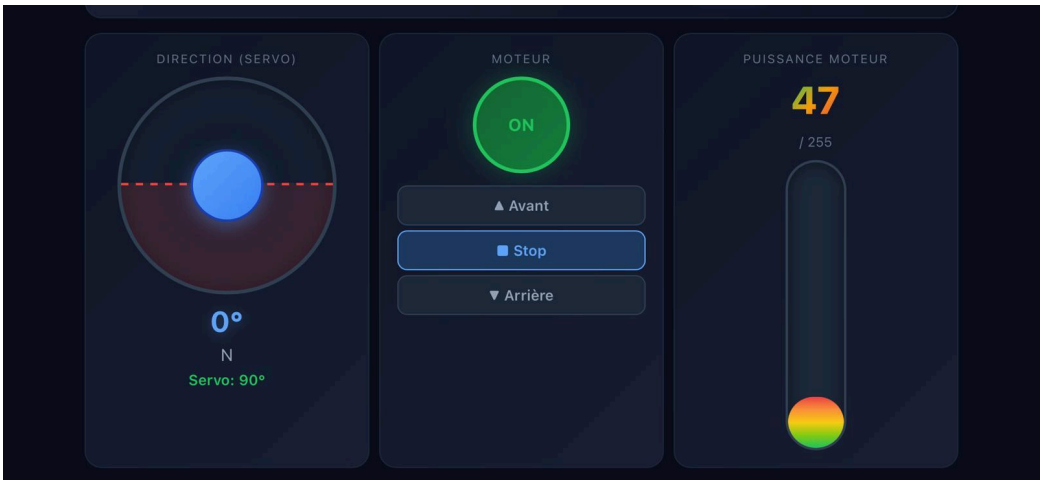
- **Objective:** Establish reliable communication
- **Implementation:**
    - Server listens for incoming HTTP requests
    - Client sends data periodically

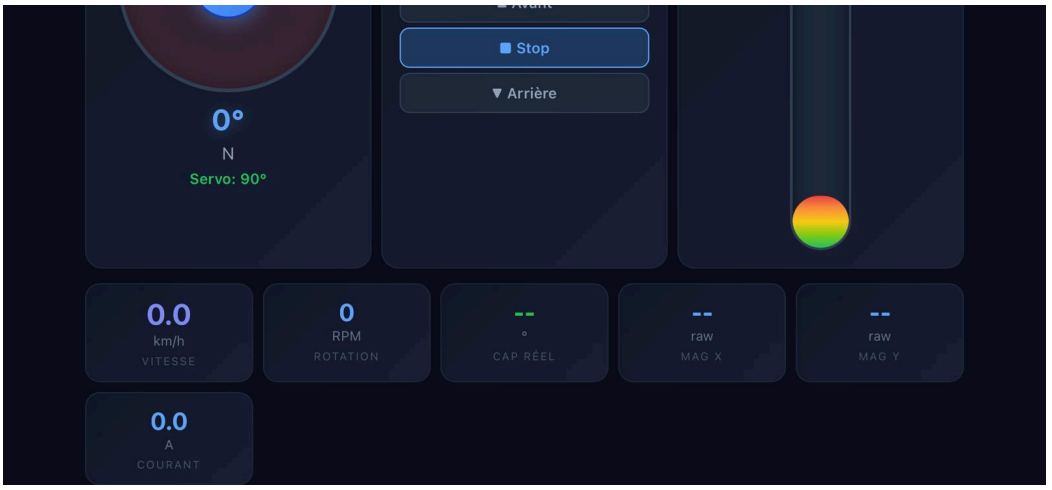
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6. Test Results and Analysis

For each test:

- **Measured Results:** Values vary but functional
- **Interface Result**





- **Analysis:**
    - The expectations needed are all good, We have good values of current, speed, motors turn forward and reversed sense, speed rotation can be controlled(0-255)
    - Had some limitations with the magnetometer reading but going on that.

7. Conclusion

The work is 85% achieved, most of the objectives are reached. Also have some key takeaways, really had difficulty in finding the relations to have calculations and value of the current using the current sensor. Also in choosing the right wires for connections.

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